



What Is a Variable?

Day 1 of 20 · Duration: 50 minutes · Theme: Minnesota State Fair

MINNESOTA STANDARDS (MDE 2022)

Standard: 9.3.5.3

Apply knowledge of number systems extending from whole numbers to integers, from integers to rational numbers, from rational numbers to real numbers and from real numbers to complex numbers to solve equations. (MP2, MP7)

Standard: 9.3.5.7

Use the structure of an expression, equation and/or formula to create an equivalent form that is more helpful given the situation. Rearrange formulas to highlight a quantity of interest, using the same reasoning in solving equations. (MP6)

Standard: 9.3.5.8

Use the structure of an expression to write it in multiple ways.

ESSENTIAL QUESTIONS

- How does a variable represent an unknown or changing quantity?
- How do we translate words into algebraic expressions?
- Why do mathematicians use letters instead of numbers?

VOCABULARY

Variable — A symbol, usually a letter, that represents an unknown or changing quantity.

Expression — A mathematical phrase made up of numbers, variables, and operation symbols (no equals sign). **Coefficient** — The number multiplied by a variable in a term (e.g., 3 in $3n$). **Constant**

— A term that is a number on its own, with no variable. **Evaluate** — To find the value of an expression by replacing variables with numbers and simplifying.

LEARNING OBJECTIVES & SUCCESS CRITERIA

- I can identify what a variable is and explain how it represents an unknown quantity.
- I can show my work step by step and check my answers for reasonableness.
- I can explain my reasoning to a partner using math vocabulary.

ENGAGEMENT STRATEGY

Strategy: Think-Pair-Share

Practice Format: Merit Time Game — Translation Match

Expectation: Every student responds on a whiteboard or to a partner before any answer is shared.

BELL RINGER — WHITEBOARDS READY!

Prompt: Write the expression for “a number increased by 6” and hold up your board. *Students write their response on a whiteboard and hold it up. Teacher scans for misconceptions before moving on.*

LESSON PACING (50 MINUTES)

Phase 1: Bell Ringer 5 min min

Write the expression for “a number increased by 6” and hold up your board.

Phase 2: Direct Instruction 10 min min

What Is a Variable? — teacher-led notes (I Do); GEMDAS reference as needed.

Phase 3: Guided Practice 12 min min

Work together (We Do) — Merit Time Game — Translation Match. Solve together: A number n is tripled then increased by 4. Write an algebraic expression. -> Answer: $3n + 4$.

Phase 4: Culture Connection 8 min min

Ojibwe [?] (Tribal Nations) — Ojibwe families harvest wild rice (manoomin) together from canoes, measuring and counting (with math word problem)

Phase 5: Practice & Game 10 min min

GO: A number n is tripled then increased by 4. Write an algebraic expression. | SET: x decreased by 9. The product of 6 and a number y . | EXTRA: Twice a number increased by 3 equals 15. Write an equation and find the number.

Phase 6: Exit Ticket 5 min min

Write an expression for “a number divided by 4, then increased by 7.”

PRACTICE PROBLEMS (GO / SET / EXTRA)

GO: A number n is tripled then increased by 4. Write an algebraic expression. **Answer:** $3n + 4$

SET: x decreased by 9. The product of 6 and a number y . **Answer:** $x - 9$ and $6y$

EXTRA: Twice a number increased by 3 equals 15. Write an equation and find the number. **Answer:** $2n + 3 = 15$, $n = 6$

CULTURE CONNECTION — MINNESOTA STORIES

Spotlight: Ojibwe [?] (Tribal Nations)

Ojibwe families harvest wild rice (manoomin) together from canoes, measuring and counting each harvest's pounds of grain.

Math Word Problem: An Ojibwe elder counts $4x$ fish in one net and $3x$ fish in another net. Write an expression for the total number of fish. If $x = 7$ fish per net, how many fish are there total?

Language fact: Ojibwemowin (Ojibwe): bezhig, niizh, niswi, niiwin (1-4)

CAREER SPOTLIGHT — MINNESOTA

Career: Mayo Clinic — Rochester | Medical Lab Scientist | BS | \$63,150 MN DEED 25th-percentile wage; OES-cited. Connection: algebra skills used daily on the job.

DIFFERENTIATION

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- Scaffolding: sentence starters for translation tasks; GEMDAS reference card on desks.
 - Extension: students write and solve their own word problems using the lesson's operation keywords.
 - Language support: bilingual vocabulary cards; pair students with a math-language peer.
 - Grouping: flexible pairs/trios by readiness; Merit Time games rotate roles each round.

ASSESSMENT

- Formative: whiteboard checks during the bell ringer; circulation during GO/SET/EXTRA; exit ticket each lesson.
- Game-based: Merit Time games give instant fluency feedback (L1–L5).
- Summative: Quiz 1 (Day 7, L1–L5), Quiz 2 (Day 17, L6–L15), End-of-Unit Test (Day 20, L1–L17).